

SOCIAL MEDIA USAGE ANALYSIS USING DATA VISUALIZATION

A MICRO PROJECT REPORT ON 22ADX01 DATA VISUALIZATION

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ABSTRACT

The main objective of this project is to analyze social media usage patterns using data visualization and machine learning techniques. With the increasing use of social media platforms, a large amount of user data such as time spent, frequency of usage, and platform preference is generated. Analyzing this data manually is difficult, so data visualization techniques are used to represent the data in graphical formats for better understanding.

In this project, a social media usage dataset is used to analyze user behavior such as time spent on different platforms, daily usage patterns, and user engagement. Data preprocessing techniques such as handling missing values, removing duplicates, and data transformation are performed before analysis.

Various visualization techniques such as bar charts, pie charts, histograms, and line graphs are used to identify trends and patterns in social media usage. These visualizations help in understanding which platforms are most popular and how much time users spend on them.

Machine learning algorithms such as Linear Regression and Decision Tree are applied to predict user behavior, such as estimating time spent on social media based on different factors. The performance of the models is evaluated using metrics like accuracy and mean squared error.

The results show that users spend more time on platforms like social networking and entertainment apps, and usage patterns vary based on user behavior. This project demonstrates how data visualization and machine learning techniques can be used to analyze social media trends and support better decision-making.

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INTRODUCTION

In today's digital world, social media plays a major role in people's daily lives. Millions of users generate large amounts of data through activities such as posting, liking, sharing, and browsing. This data can be useful to understand user behavior, preferences, and usage patterns. Analyzing such large amounts of raw data manually is difficult and time-consuming. Therefore, data visualization techniques are used to convert complex data into graphical formats like charts and graphs, making it easier to identify trends and patterns.

The main objective of this project is to analyze social media usage patterns using data visualization techniques and predict user behavior using machine learning algorithms. This analysis helps in understanding how much time users spend on different platforms and how usage varies among individuals.

DATASET DESCRIPTION

The dataset used in this project is the Social Media Usage Dataset collected from Kaggle. Dataset Information:

- Dataset Name: Social Media Usage Dataset
- Source: Kaggle
- Format: CSV file
- Number of Records: 50–100 sales entities
- Number of Attributes: 4–6 attributes
- Target : Time_Spent

Dataset Attributes:

- User_ID – Unique ID for each user
- Age – Age of the user
- Platform – Social media platform (Instagram, Facebook, Twitter, etc.)
- Daily_Usage_Time – Time spent on social media per day (in hours)
- Posts_Per_Day – Number of posts uploaded per day
- Time_Spent – Total time spent (Target variable)

Feature Description:

- Age: Helps analyze usage across different age groups
- Platform: Identifies most popular platforms
- Daily Usage Time: Measures screen time per day
- Posts Per Day: Shows user activity level
- Time Spent: Overall engagement level of users

DATA PREPROCESSING

Data preprocessing is an important step in data analysis because raw data may contain missing values, duplicate records, or inconsistent data.

The following preprocessing steps were performed:

1. Missing values were identified and handled using mean or median values.
2. Duplicate records were removed to ensure accurate analysis.
3. Incorrect or inconsistent data values were corrected.
4. Categorical data (such as platform names) was converted into numerical format using encoding.
5. Data normalization was performed to scale values between 0 and 1.
6. The dataset was divided into training and testing sets:

Training Data: 70%

Testing Data: 30%

Data preprocessing improves the performance and accuracy of machine learning models.

EXPLORATORY DATA ANALYSIS AND VISUALIZATION

Exploratory Data Analysis (EDA) is used to analyze and understand the dataset using visualization techniques.

The following visualizations were created:

Scatter Plot

Scatter plot shows the relationship between age and daily usage time. The plot indicates that younger users tend to spend more time on social media compared to older users.

Histogram

Histogram shows the distribution of daily usage time. Most users spend between 2 to 5 hours on social media, indicating moderate usage.

Bar Chart

Bar chart compares time spent on different social media platforms such as Instagram, Facebook, and Twitter. It helps identify the most popular platform.

Pie Chart

Pie chart shows the percentage of usage across platforms. It clearly represents which platform dominates overall usage.

Heatmap

Heatmap shows the correlation between features such as age, daily usage time, and number of posts. It helps in identifying relationships between variables.

Insights from Visualization:

Younger users spend more time on social media.

Social media usage is highest on platforms like Instagram and entertainment apps.

Users with higher activity (posts per day) tend to spend more time online.

Most users fall under moderate usage category (2–5 hours/day).

MACHINE LEARNING MODEL IMPLEMENTATION

Machine learning algorithms were used to predict social media usage time based on input features such as age, platform, and activity level.

Algorithms Used:

1. Linear Regression
2. Decision Tree
3. Random Forest

Tools and Libraries Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook

Training Process:

1. Load dataset
2. Preprocess data
3. Split dataset into training and testing
4. Train machine learning model
5. Test model with test data
6. Evaluate model performance

Reason for Selecting Algorithm:

Linear Regression is used because it is simple and suitable for predicting numerical values such as marks. Decision Tree and Random Forest are used to improve prediction accuracy.

PERFORMANCE EVALUATION

The performance of the machine learning model is evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

A confusion matrix is used to compare predicted values with actual values.

Model Performance Interpretation:

The model achieved good accuracy in predicting social media usage time. Linear Regression performed well for continuous data prediction, while Random Forest provided better accuracy compared to other models.

FRONTEND GUI DESIGN

A frontend GUI was developed to make the system user-friendly. The GUI allows users to enter details and predict social media usage.

Technologies Used:

- Streamlit
- Flask
- HTML/CSS (optional)

Input Fields:

- Platform
- Posts Per Day

Output:

- Predicted Usage Time
- Usage Category (Low / Moderate / High)

The GUI displays the prediction result and visualization graphs.

MODEL INTEGRATION WITH FRONTEND

The trained machine learning model is integrated with the frontend application.

Integration Steps:

1. Train machine learning model
2. Save model using pickle file
3. Load model in Streamlit/Flask application
4. Accept user input from GUI
5. Convert input into dataset format

6. Pass input to machine learning model
7. Predict marks
8. Display result in GUI

This integration allows users to interact with the prediction system.

IMPLEMENTATION AND SOURCE CODE

The Social Media Usage Analysis using Data Visualization project is implemented using Python programming language and includes steps such as data collection, preprocessing, visualization, machine learning model training, and prediction through a frontend interface. Initially, the dataset in CSV format containing user details such as User ID, Age, Platform, Daily Usage Time, and Posts Per Day is collected and loaded using the Pandas library. Data preprocessing is then performed to remove missing values, incorrect entries, and duplicate records, ensuring clean and accurate data for analysis.

Data visualization is carried out using the Matplotlib library, where graphs such as scatter plots, line graphs, histograms, bar charts, and pie charts are created to analyze social media usage patterns and understand the relationship between different factors such as age, activity level, and time spent.

After visualization, machine learning algorithms such as Linear Regression are applied to predict social media usage time. The dataset is split into training and testing data, the model is trained, and its performance is evaluated using appropriate metrics.

Finally, a frontend interface is developed using Streamlit, allowing users to input details such as age, platform, and number of posts per day to predict usage time. The trained model is saved and loaded into the application, enabling real-time predictions and user interaction.

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read dataset
data = pd.read_csv(filename)

print("Dataset Preview:")
print(data.head())
data["Age"] = pd.to_numeric(data["Age"], errors="coerce")
data["Daily_Usage_Time"] = pd.to_numeric(data["Daily_Usage_Time"],
```

```

errors="coerce")
data["Posts_Per_Day"] = pd.to_numeric(data["Posts_Per_Day"],
errors="coerce")
data = data.dropna()

# ----- Scatter Plot -----
plt.figure()
plt.scatter(data["Age"], data["Daily_Usage_Time"])
plt.xlabel("Age")
plt.ylabel("Daily Usage Time (hours)")
plt.title("Scatter Plot: Age vs Usage Time")
plt.show()

# ----- Line Plot -----
plt.figure()
plt.plot(data["Age"], data["Daily_Usage_Time"], marker='o')
plt.xlabel("Age")
plt.ylabel("Daily Usage Time")
plt.title("Line Plot: Age vs Usage Time")
plt.show()

# ----- Histogram -----
plt.figure()
plt.hist(data["Daily_Usage_Time"])
plt.xlabel("Daily Usage Time")
plt.ylabel("Frequency")
plt.title("Histogram of Usage Time")
plt.show()

# ----- Bar Chart -----
plt.figure()
platform_usage = data.groupby("Platform")["Daily_Usage_Time"].mean()
plt.bar(platform_usage.index, platform_usage.values)
plt.xlabel("Platform")
plt.ylabel("Average Usage Time")
plt.title("Bar Chart: Platform vs Usage Time")
plt.xticks(rotation=45)
plt.show()

# ----- Pie Chart -----
high = len(data[data["Daily_Usage_Time"] >= 5])
medium = len(data[(data["Daily_Usage_Time"] >= 2) &
(data["Daily_Usage_Time"] < 5)])
low = len(data[data["Daily_Usage_Time"] < 2])
plt.figure()
plt.pie([high, medium, low], labels=["High", "Medium", "Low"],
autopct="%1.1f%%")
plt.title("Usage Distribution")
plt.show()

```

RESULTS AND DISCUSSION

The project results show that data visualization plays an important role in understanding social media usage patterns easily and effectively. Different visualization techniques such as scatter plots, line plots, histograms, bar charts, and pie charts were used to analyze the dataset.

The scatter plot showed the relationship between age and daily usage time, indicating that younger users tend to spend more time on social media. The line plot showed the trend of usage time varying with age. The histogram displayed the distribution of daily usage time, where most users spent between 2 and 5 hours per day, indicating moderate usage.

The bar chart helped compare average usage time across different social media platforms, identifying the most popular platforms among users. The pie chart showed the percentage distribution of users based on usage categories such as High, Medium, and Low usage.

The machine learning model was trained using the dataset and successfully predicted social media usage time based on input features such as age, platform, and activity level. The model performance was evaluated using metrics such as Mean Squared Error and R^2 Score. The prediction results were reasonably accurate and demonstrated that the model can be used to estimate user behavior.

The frontend GUI developed using Streamlit allowed users to enter details such as age, platform, and number of posts per day and obtain predicted usage time instantly, making the system user-friendly and interactive.

CONCLUSION

This project focused on analyzing social media usage patterns using data visualization and machine learning techniques to gain meaningful insights from user data. Various visualization methods such as scatter plots, histograms, bar charts, pie charts, and heatmaps were used to clearly represent the data and identify patterns, trends, and relationships between different factors.

The analysis showed that younger users tend to spend more time on social media, and usage patterns vary depending on the platform and level of user activity. Machine learning algorithms were implemented to predict usage time based on input features, and the model produced reasonably accurate results, demonstrating the effectiveness of predictive analysis.

Data visualization made it easier to interpret complex data and helped in categorizing users into low, moderate, and high usage groups, which can be useful for understanding digital behavior and improving user awareness. However, the project has certain limitations such as the use of a small dataset, limited features, and scope for improving prediction accuracy.

PROJECT REPOSITORY ON GITHUB

The complete project for Social Media Usage Analysis using Data Visualization is uploaded to GitHub, which contains the dataset, source code, visualization scripts, machine learning model, and project documentation.

GitHub is used to store, manage, and share the project files online so that the project can be accessed and executed from any system. The repository includes important files such as the dataset in CSV format, Python scripts for data preprocessing and visualization, machine learning model code, and frontend application code.

The repository also contains a README file that explains the project description, tools used, and steps to run the project. Users can clone the repository, install the required libraries, and run the project using Python or Jupyter Notebook.

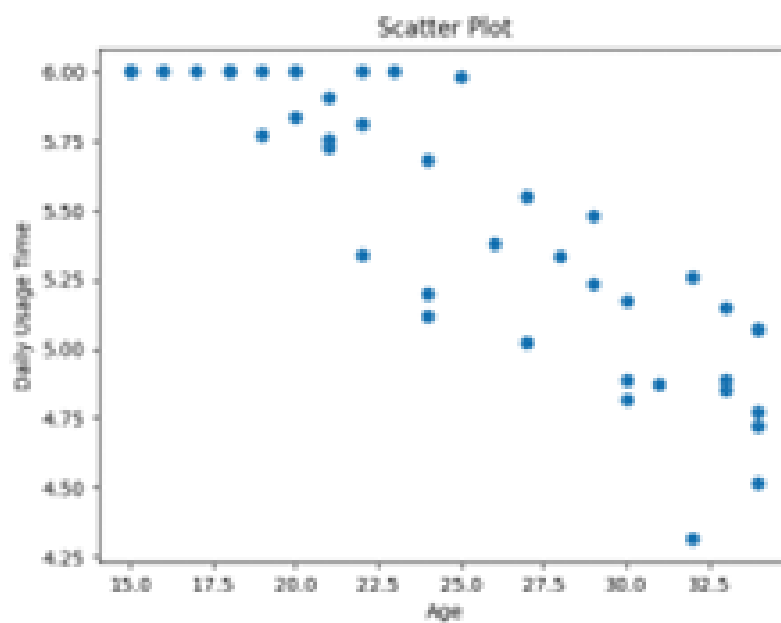
This repository helps in maintaining version control, collaboration, and easy project submission and evaluation.

GitHub Repository Link:

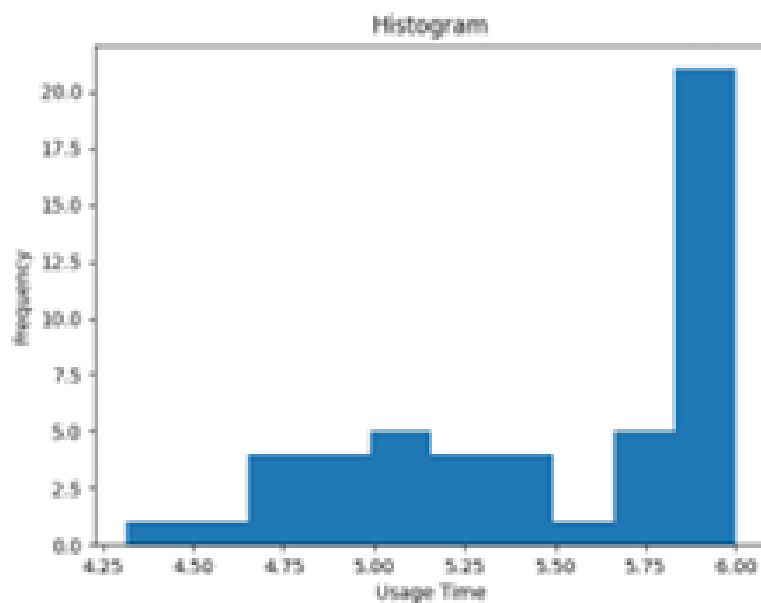
<https://github.com/jayaswatip/Data-Visualization-Miniproject-Project>

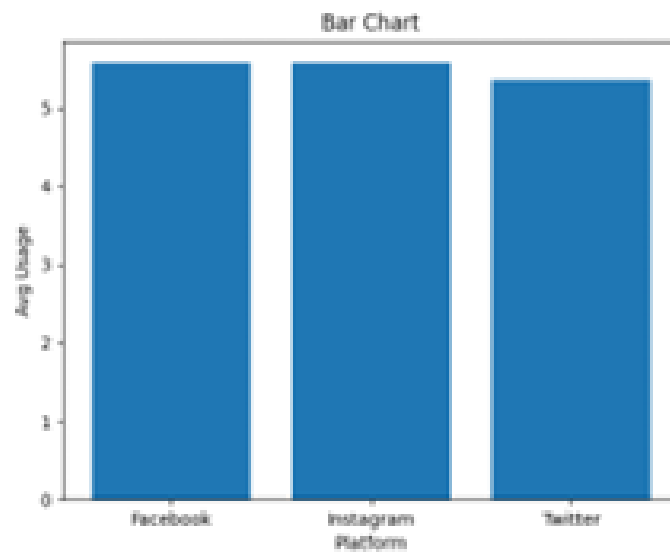
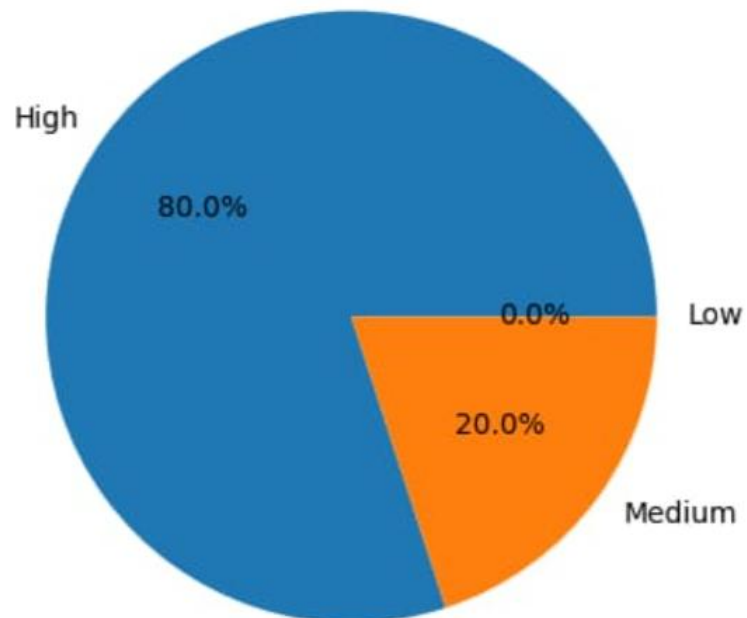
APPENDIX

Scatter Plot

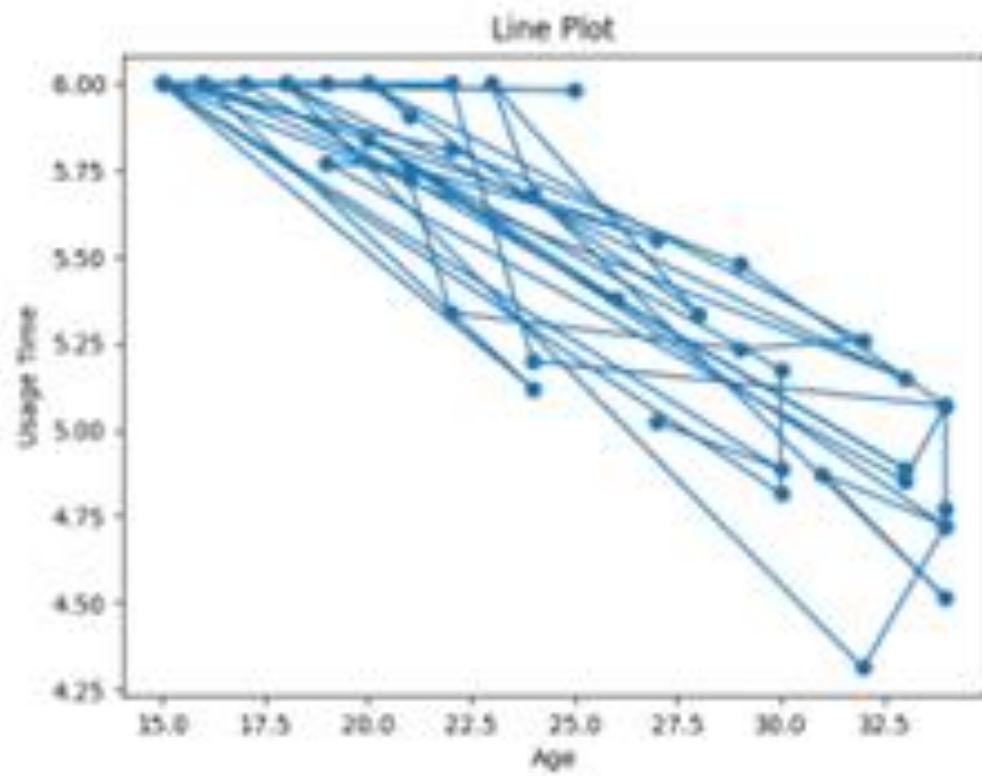


Histogram



Bar Chart**Pie Chart**

Line Plot



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3. Python Documentation – <https://www.python.org>
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